

MAHMUD HASAN RATUL

Full Stack Engineer | AI Researcher

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Computer Science & Engineering student with production-level Full Stack and AI engineering experience. Built applications used by 10,000+ users, deployed platforms serving 1K+ concurrent users, and grew a university tech community from 200 to 500+ members. Strong in end-to-end product development — from system architecture and LLM integration to event leadership and team management at national scale.

EXPERIENCE

Vice President (DIU CPC) Jan 2025 – Present

Led strategic operations, technical initiatives, and community development for one of Bangladesh's largest university computer science clubs -DIU Computer & Programming Club (DIUCPC). Managed a 27-member core team, expanded executive membership from 200 to 500+ students, and organized 15+ programming contests, hackathons, and technical workshops while mentoring and guiding junior developers.

Co-Lead Volunteer (ICPC) — Crowd Control & Operations Head Dec 2024

Organizing Volunteer, ICPC Dhaka Regional Contest 2024 — Coordinated arena operations, volunteer management, and contest-day logistics for 100+ participating universities. Managed on-site technical support, contestant assistance, venue allocation, and operational workflows to ensure the smooth execution of a high-density regional programming competition.

Head of Branding & Production(HULT PRIZE) 2025

Hult Prize at Daffodil International University — Led the creative direction for event branding, including promotional videos, graphics, and media campaigns. Coordinated production timelines and managed cross-functional creative workflows to ensure consistent and impactful event communication.

Marketing Executive(TPHL) Nov 2024 – 2025

Marketing & Branding Executive, The Premium Homes Ltd. — Developed real estate branding materials and executed digital marketing campaigns for premium property listings. Managed promotional content, outreach initiatives, and client-facing communications to enhance brand visibility and engagement.

TECHNICAL SKILLS

Languages: C, Java, JavaScript (ES6+), TypeScript, Python

Frontend: React.js, Next.js, React Native, Vite, Redux, Tailwind CSS, HTML5/CSS3

Backend: Node.js, Express.js, FastAPI, Socket.io, REST APIs

Databases: MongoDB, PostgreSQL, Firebase (Realtime + Firestore), Redis

AI / ML: Gemini API, OpenAI GPT-4, Whisper, TensorFlow, PyTorch, LLM Prompt Engineering, Agentic Flows

DevOps: Docker, GitHub Actions (CI/CD), Vercel, Git, Linux, Shell Scripting

Tools: MS Office Suite, Google Workspace, Canva, Video Editing, Digital Photography

Soft Skills: Technical Leadership, Event Logistics, Public Speaking, Project Management, Mentoring

PROJECTS

CV Maker AI | Next.js 14, Gemini Pro API, Stripe, TypeScript, Tailwind CSS

- LLM-powered ATS resume analyzer and builder; scans resumes against target job roles and generates actionable improvements.
- Achieved 92% ATS accuracy scoring and ~15ms API response latency; serving 500+ active users.

Orivo Commerce | Next.js 14, Firebase, Redux, Stripe SDK, Redis, PWA

- Enterprise e-commerce platform with real-time inventory sync, secure Stripe checkout, and Redis-backed session management.
- Architected for 10K+ concurrent users; validated at 99.9% uptime in production.

Lecture AI | React, OpenAI Whisper, GPT-4, FastAPI, PostgreSQL

- End-to-end NLP pipeline: audio upload → Whisper transcription → GPT-4 summarisation → structured markdown output.
- Transcribed 100+ hours of educational audio with 92% accuracy; deployed via FastAPI on cloud infrastructure.

NeuroStack | React 18, Firebase, Gemini 1.5 Flash API, Tailwind CSS, PWA

- PWA that auto-fetches daily RSS feeds and YouTube content, runs Gemini summarisation, and stores entries in a personal knowledge dashboard.
- Fully offline-cached; zero recurring API cost overhead through smart quota management.

UniVibe | React 18, TypeScript, Vite, ESLint

- Production-grade web starter with instant HMR, strict TypeScript linting, and clean scalable project architecture.
- Adopted by 300+ developers as their preferred React/TypeScript boilerplate.

EDUCATION

Bachelor of Science in Computer Science & Engineering Spring 2023 – Present

Daffodil International University (DIU), Dhaka, Bangladesh

Higher Secondary Certificate (HSC) — Science 2021

Ghatail Cantonment Public School & College, Tangail — GPA: 5.00 / 5.00

Secondary School Certificate (SSC) — Science 2019

Suti V.M. Pilot Model Govt. High School, Tangail — GPA: 4.94 / 5.00

EXTRACURRICULAR & VOLUNTEERING

Volunteer — Bangladesh Olympiad in Informatics (BDOI) 2024 and 2025

Campus Ambassador — Jobsnavi 2024

Conference Volunteer — IDAA 2025 (International Conference on Intelligent Data Analysis and Applications) 2025

Conference Volunteer — BIM 2023 (Big Data, IoT & Machine Learning) 2023

General Member — DIU Blood Donors Club (DIU BDC) 2023 – Present

REFERENCES

Md. Mainul Hasan Dulon

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Mr. Mayen Uddin Mojumdar

Assistant Professor, Dept. of CSE — DIU

Academic Reference & Mentor

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